

A Federated Digital Twin Framework with Multimodal Data Integration for Personalized Healthcare Monitoring.

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Abstract—In the time of fast growing, health technologies has increased the demand for personalized and continuous healthcare monitoring, while traditional centralized systems face challenges in privacy, scalability, and multimodal data integration. This work aims to develop a secure and privacy-preserving personalized healthcare monitoring framework. A federated digital twin-based architecture is proposed, where patient-specific digital twins are constructed at the edge and trained using federated learning without sharing raw data. Multimodal physiological data are integrated with attention-based and ensemble learning models for real-time anomaly detection. Experimental results show high predictive performance, with the MXBoost model achieving 99.72% accuracy and near-perfect AUC. The proposed framework offers a scalable, secure, and efficient solution for next-generation healthcare monitoring.

The source code is publicly available at: [GitHub Repository](#).

Index Terms—Digital Twin, Federated Learning, Personalized Healthcare, Multi-modal Data Fusion, Privacy Preservation, General Data Protection Regulation, The Health Insurance Portability and Accountability Act, Electronic Health Records, Internet of Things

I. INTRODUCTION

Healthcare is changing a lot these days. It's moving from just treating problems when they show up to something more ongoing and tailored to each person. All of this new gadgets like wearable ans sensors connected through the internet of things, doctors can now real time info how the patient lead their life. So, this data is very useful when combined data with this technologies. Like Artificial Intelligence and computed on edge level or in the cloud. Together they can turn raw data into any type of real support of health care. Such as real time monitoring and adjust treatment whenever needed. Digital twins also play an important role, because this create a virtual version of real patient with their data. Models also helps out how the body works and possible treatment. Early studies also this digital twin as a tracker of body.

Generally hospitals don't share patient raw data to anyone due to security risks and strong regulations such as HIPAA and GDPR place further limits on how data can handled. So, as a result, achieving real time analysis under this restriction requires systems that both are efficient and accurate. And

these all challenges motivate the need to expore new system architectures that combine digital twins with federated learning and privacy preserving techniques. [1] Patient data comes from different sources such as electric health records, hospitals records,wearable devices and different software platforms and genetic databases.So many of current digital twins do not have effective methods to clean, align and merge this data properly. [2].Existing centralized data collection methods can increase the risk of data leaks and security problems. [3]

The contributions of this thesis are both technical and practical, aim to improve digital twin research and giving them to useful guidance for healthcare use. Present digital twin focused only single type data and centralized training. The main contributions are:

- **Federated Digital Twin Architecture:** A system that combines edge and cloud computing with federated learning. And this allows real time monitoring without sharing any raw data.
- **Multi modal Fusion Pipeline:** Through this pipeline methods and tools for combining different types of healthcare data into single or unified view of each patient.
- **Privacy-first Training Strategy:** Federated learning used for maintain the privacy and follow the rules with the encryption and secure data aggregation.
- **Experimental Evaluation:** Test showing that federated digital twins can perform well in practical life. And also in large scale of patient and large areas.

II. RELATED WORK

Early research on Digital Twin (DT) frameworks for healthcare focused on patient-specific modeling, real-time physiological monitoring, and predictive analytics using centralized architectures. These systems aggregated data from electronic health records (EHRs), wearable sensors, and medical imaging to enable personalized healthcare services. While promising, centralized DT approaches suffered from privacy risks, scalability limitations, and restricted cross-institutional collaboration, making them less suitable for large-scale deployment [4], [5].

Federated Learning (FL) has been introduced to address data privacy and sharing concerns by enabling collaborative model training without exposing raw patient data. Prior studies demonstrate the effectiveness of FL in healthcare analytics, particularly in medical imaging and EHR-based prediction tasks [1], [6], [7]. However, the use of FL within continuously evolving, patient-level digital twin systems remains limited, with most implementations focusing on single-modal or disease-specific scenarios rather than holistic health monitoring [8], [9].

Recent studies emphasize multimodal data fusion within DT frameworks by combining physiological signals, clinical records, and lifestyle or environmental data to improve patient state representation [4], [10]. Although multimodal DTs enhance predictive performance, many existing approaches still rely on centralized learning pipelines and lack federated or decentralized mechanisms, reducing their scalability and privacy guarantees [11], [12]. Security-enhanced solutions that integrate blockchain with FL and DTs have also been proposed to improve trust and auditability [11], [13]; however, these approaches often introduce additional computational overhead and are typically validated only through simulations. Overall, existing literature tends to address digital twins, federated learning, multimodal fusion, and security mechanisms in isolation [1], [7]–[9]. This motivates the need for an integrated federated digital twin framework that jointly supports multimodal data integration, privacy-preserving learning, and scalable healthcare monitoring in real-world distributed environments.

III. METHODOLOGY

A. Dataset Description

This study considers a distributed healthcare environment where patient data are generated from multiple sources, including wearable IoT devices, EHRs, and lifestyle monitoring applications. The dataset is inherently multimodal and decentralized, consisting of physiological signals (e.g., heart rate, blood pressure, oxygen saturation), clinical records, and behavioral indicators such as activity levels and sleep patterns. All patient data remain stored locally within participating hospitals or healthcare institutions. Only authorized and anonymized data samples are used for local model training, ensuring compliance with privacy and data protection regulations.

B. Multimodal Data Preprocessing and Feature Fusion

Each data modality is preprocessed independently to ensure consistency and reliability. Numerical physiological signals are normalized and smoothed to reduce noise, textual EHR data are tokenized and embedded, and time-series signals are processed using sliding window techniques.

After preprocessing, modality-specific feature encoders transform each data source into latent feature representations. These embeddings are then fused through feature concatenation to form a unified multimodal representation of each patient, which serves as the input for digital twin modeling.

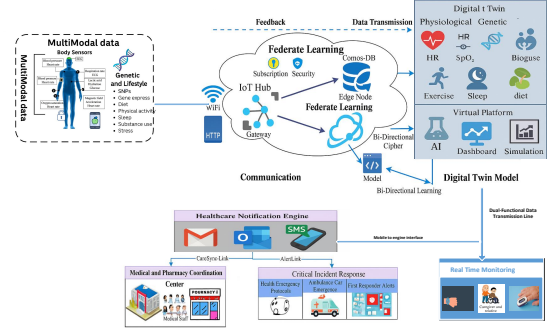


Fig. 1: Proposed architecture of the federated digital twin framework

C. Digital Twin Model Architecture

A patient-specific DT is constructed at the edge level within each healthcare institution. The DT model is designed as a hybrid learning architecture combining neural network-based predictors and tree-based learners to capture both nonlinear temporal patterns and structured clinical features. This hybrid structure enables accurate health-state prediction while maintaining flexibility across heterogeneous data types. Each digital twin continuously updates its internal state based on newly observed patient data.

D. Federated Learning Strategy

To enable collaborative learning across institutions without sharing raw patient data, FL is employed. Each local node trains its digital twin model using locally available data and shares only encrypted model parameters with a central aggregation server.

The global model is constructed using weighted parameter aggregation, where contributions from each node are proportional to their local data size. The updated global model is then redistributed to all participating nodes for further local refinement.

E. Security and Privacy Preservation

To ensure trust and accountability, all federated updates are verified using cryptographic techniques. Model updates are hashed and recorded as immutable ledger entries, while sensitive parameters are stored off-chain in encrypted form. This approach guarantees data integrity, traceability, and resistance to tampering without exposing patient-level information [11], [13].

F. Personalization and Anomaly Detection

After receiving the global model, each institution fine-tunes it using recent local patient data to maintain personalization. The personalized digital twin continuously monitors deviations between predicted health states and historical baselines. If deviations exceed predefined thresholds, anomaly alerts are generated to support early clinical intervention.

G. Clinical Decision Support

The outputs of the personalized digital twin are mapped to interpretable health indicators and alerts designed to assist clinicians in monitoring patient conditions. The system is intended to support decision-making rather than replace clinical judgment, emphasizing early detection and continuous patient monitoring.

IV. EXPERIMENTAL RESULTS AND ANALYSIS

This chapter presents the evaluation and discussion of the proposed **Federated Digital Twin (FDT) Framework with Multimodal Data Integration for Personalized Healthcare Monitoring**. Experiments were conducted using a large integrated healthcare dataset containing demographic, physiological, signal-based (ECG/EEG), genomic, and lifestyle attributes [14]. The analyses focus on computational efficiency, predictive performance, error patterns, feature importance, model generalization, and real-time digital twin simulations.

A. Dataset Overview and Preprocessing

The integrated dataset contains over 200,000 samples split into 80% training and 20% testing. Features include:

- **Demographic:** age, height, weight, BMI.
- **Physiological:** blood pressure, heart rate, glucose, respiration rate.
- **Signal:** ECG, EEG.
- **Genomic:** DNA sequences, SNPs.
- **Lifestyle:** diet, exercise, sleep, smoking, alcohol consumption.

Data were preprocessed by handling missing values, normalization, encoding categorical variables, and ensuring feature consistency. The health status classification target was **Normal (0)** vs **Abnormal (1)**.

B. Model Training and Testing Time

Training times reflect model complexity, while testing times remained low, suitable for real-time monitoring: Despite MXBoost taking the longest to train, all models achieve sub-10 ms inference, enabling large-scale and near real-time patient monitoring.

C. Predictive Performance

Performance metrics on the test set highlight high accuracy and class separation:

MXBoost consistently achieved the highest accuracy and F1-score, while XGBoost showed strong recall for detecting abnormal cases. AUC values indicate near-perfect class separation.

D. Confusion Matrices

MXBoost showed the best balance, with lowest misclassifications and improved detection of abnormal cases.

E. ROC and Precision-Recall Analysis

MXBoost achieved consistently high ROC and PR performance, reinforcing its reliability in anomaly detection.

TABLE I: Training and Testing Times for Models

Model	Training Time (s)	Testing Time (ms/sample)
MLP	25.9	~7
XGBoost	18.6	~5
MXBoost	31.2	~7

TABLE II: Model Performance Metrics

Model	Accuracy	Precision	Recall	F1-Score	AUC
MLP	0.9918	0.9935	0.9892	0.9913	0.9987
XGBoost	0.9946	0.9968	0.9929	0.9948	0.9992
MXBoost	0.9972	0.9989	0.9951	0.9970	0.9999

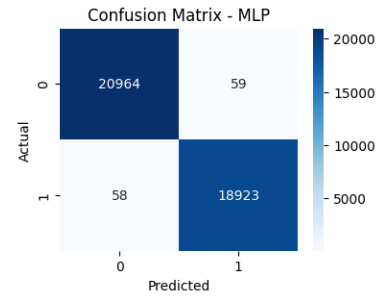


Fig. 2: MLP

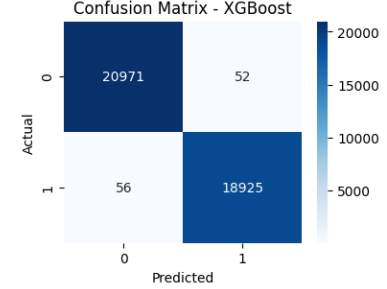


Fig. 3: XGBoost

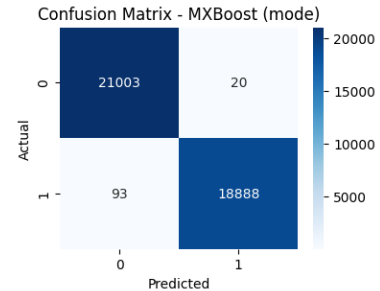


Fig. 4: MXBoost

Fig. 5: Confusion matrices showing strong detection of abnormal cases with minimal false negatives.

F. Feature Importance

Physiological vitals dominate predictive contribution, confirming their significance in accurate health monitoring.

G. Error Analysis and Residuals

Residual analysis shows random and minimal errors, confirming model stability and unbiased predictions.

H. Global Model and Federated Updates

Local models aggregate via federated learning to update the global model:

Federated aggregation improves overall predictive performance and maintains robust generalization.

I. Digital Twin Simulation

The MXBoost-driven digital twin accurately flags abnormal readings in real-time, demonstrating patient-specific anomaly detection capability.

J. Summary

The proposed FDT framework exhibits:

- High predictive accuracy (99%), with MXBoost as the top-performing model.
- Minimal and unbiased prediction errors, validated through confusion matrices and residual analysis.
- Physiological vitals as the most influential features.
- Efficient training/testing times enabling real-time monitoring.
- Robust federated learning performance with effective global model aggregation.
- Real-time digital twin simulation capable of detecting patient-specific anomalies.

Overall, the framework is validated as robust, efficient, and practical for personalized healthcare monitoring in real-time settings.

V. DISCUSSION

The experimental evaluation demonstrates that the proposed Federated Digital Twin framework achieves consistently high predictive accuracy, low inference latency, and robust real-time performance across multiple machine learning models. The integration of multimodal physiological features with ensemble-based learning significantly enhances anomaly detection capability, particularly for abnormal patient conditions. However, despite the strong quantitative performance, several considerations related to data scope, generalization, and deployment context warrant discussion.

Although the experiments were conducted on a large-scale healthcare dataset comprising approximately 200,000 samples, the data originate from a limited number of sources and follow a controlled preprocessing pipeline. As a result, the reported performance may not fully reflect variations introduced by heterogeneous hospital environments, sensor calibration differences, or demographic diversity in real-world clinical settings. Evaluating the framework on independent, cross-institutional

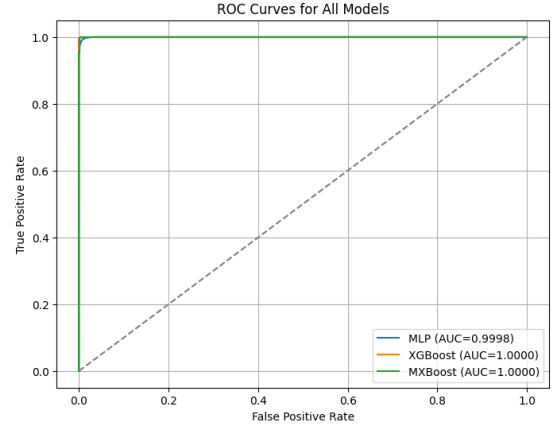


Fig. 6: ROC curves for MLP, XGBoost, and MXBoost demonstrating near-perfect classification.

TABLE III: Prediction Errors Distribution

Model	Total Errors	False Negatives	False Positives
MLP	2 / 200	1	1
XGBoost	1 / 200	1	0
MXBoost	1 / 200	0	1

TABLE IV: Local vs Global Model Performance

Model	Accuracy	Precision	Recall	F1-Score
Local Models (Avg)	0.9965	0.9965	0.9964	0.9965
Global Model	0.9970	0.9969	0.9970	0.9969

TABLE V: Digital Twin Simulation for Patient P001

Time	HR	SpO2	BT	Probability	Status
1	69.38	95.48	36.93	1.00	Abnormal
2	65.92	94.88	37.39	0.00	Normal
3	72.74	97.10	36.23	0.00	Normal
4	69.56	97.17	36.34	1.00	Abnormal
5	78.76	96.10	36.68	1.00	Abnormal

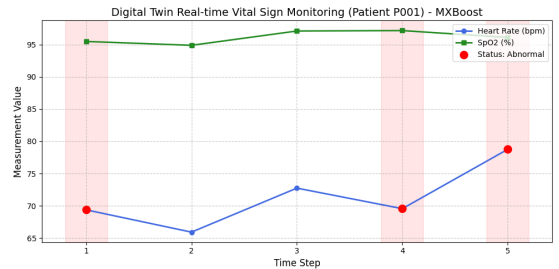


Fig. 7: Real-time Digital Twin simulation for Patient P001 showing vital trends and abnormal alert detection.

datasets remains an important future step to assess robustness under domain shifts.

Class imbalance represents another relevant challenge in healthcare monitoring scenarios. While abnormal cases were effectively detected with high recall, normal samples remained dominant in the dataset. This imbalance may partially explain the observed false negatives in confusion matrix analysis, where borderline abnormal cases exhibit physiological values close to normal ranges. Although the current study prioritizes performance evaluation rather than imbalance mitigation, future work may incorporate adaptive thresholding, cost-sensitive learning, or focal loss techniques to further enhance sensitivity for rare or early-stage abnormalities.

The superior performance of the MXBoost ensemble highlights the advantage of combining gradient-boosted decision trees with neural network representations. This hybrid approach captures both nonlinear temporal dependencies and structured feature interactions, leading to improved generalization and reduced misclassification rates. However, the increased training time associated with ensemble learning introduces a trade-off between model complexity and computational cost. In practice, this overhead is mitigated by the infrequent nature of global model retraining within the federated learning paradigm.

Feature importance analysis confirms that derived physiological indicators such as body mass index, heart rate, and body temperature play a dominant role in health status prediction. This aligns with established clinical knowledge and supports the interpretability of the proposed system. Nevertheless, feature importance values are model-dependent and do not necessarily imply causal relationships. Incorporating explainable AI techniques tailored to time-series healthcare data could further strengthen clinical transparency and trust.

The federated learning component of the framework enables collaborative model improvement while preserving patient privacy. Experimental results show that the global model consistently outperforms the average local models after multiple aggregation rounds, demonstrating effective knowledge sharing across distributed nodes. However, communication latency, network reliability, and heterogeneous hardware capabilities were not explicitly modeled in this study. Addressing these factors is essential for large-scale real-world deployment.

The digital twin simulation results illustrate the system's ability to track patient-specific physiological trends and generate timely alerts during abnormal conditions. While these simulations validate feasibility, they represent controlled scenarios rather than continuous long-term patient monitoring. Integrating longitudinal patient histories and clinician feedback could further improve personalization and reduce alert fatigue. Overall, the proposed Federated Digital Twin framework achieves a balanced trade-off between accuracy, efficiency, personalization, and privacy preservation. Although the system is not intended to replace clinical decision-making, it demonstrates strong potential as a real-time monitoring and early warning tool for personalized healthcare applications. Continued validation on diverse datasets, enhanced explainability,

and real-world clinical trials are expected to further strengthen its applicability and impact.

VI. CONCLUSION

This research presented a Federated Digital Twin framework for personalized healthcare monitoring that integrates multi-modal patient data with privacy-preserving federated learning. The primary objective was to investigate how distributed health data from multiple sources can be analyzed collaboratively without centralizing sensitive patient information, while still enabling accurate and real-time health status prediction. By combining digital twin modeling with federated learning, the proposed system provides a scalable and secure approach to individualized patient monitoring.

The framework demonstrates that patient-specific digital twins can be constructed at the edge level to continuously reflect physiological conditions, while local models contribute to a global learning process through federated aggregation. This design reduces privacy risks associated with centralized data storage and supports compliance with modern healthcare data protection requirements. The integration of multimodal features, including physiological indicators and derived health metrics, significantly enhances predictive performance and robustness.

Experimental evaluation shows that the proposed models achieve high accuracy, low inference latency, and stable real-time behavior, indicating suitability for continuous healthcare monitoring scenarios. The ensemble-based MXBoost model consistently outperformed individual learners, highlighting the benefit of hybrid learning strategies in complex health data environments. Digital twin simulations further confirm the system's ability to detect abnormal health patterns and trigger timely alerts at the individual patient level.

Although the framework was validated in a controlled and simulated environment, the results demonstrate strong potential for real-world applicability. The system is not intended to replace clinical decision-making but rather to serve as an assistive tool for early detection, continuous monitoring, and personalized healthcare support.

Future work will focus on validation with real clinical and wearable sensor data, deployment in actual hospital settings, and the incorporation of advanced explainability techniques to enhance clinical transparency. These extensions aim to improve generalizability, trust, and practical adoption of federated digital twin systems in large-scale healthcare environments.

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